

~~Verilog~~Day 2

1) Derive the output expression f for the below circuits and draw the simplified circuits

a) Gate 1: (~~XOR~~) : $AB + \overline{A}B$ or $(A \oplus B)$

b) Gate 2: (AND) : BC

c) Gate 3: (~~XOR~~) : $(AB + \overline{A}B) + B = \overline{A} + B$

Gate 4: $(\overline{A} + B) \oplus BC$

$f = \overline{A}B + \overline{A}C + BC$

b) Gate 1: (AND1) : $A \overline{B}$

Gate 2: (AND2) : $A B$

Gate 3: (OR1) : $AB + B = B$

Gate 4: (OR2) : $A \overline{B} + B = A + B$

Gate 5: (AND3) : $(A + B) B = B$

Gate 6: (OR3) : $B + A$

$f = A + B$

c) Gate 1: (NOT1) : $\overline{A}$

Gate 2: (OR1) : $\overline{A} + B$

Gate 3: (NOT2) : $\overline{(\overline{A} + B)} = A \overline{B}$

Gate 4: (~~AND1~~) : $A \overline{B} \cdot B = 0$

Gate 5: (NOT3) : $\overline{0} = 1$

Gate 6: (~~AND2~~) : $1 \cdot C = C$

Gate 7: (OR2) : $C + B$

$f = B + C$

2) Minimise the following Boolean Expressions using Boolean postulates

a) $f = \bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

Ans:

$$f = \bar{A}BC\bar{D}E + \bar{A}\bar{B}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCDE + ABC\bar{D}E + ABD$$

$$f = \bar{A}BC\bar{D}E + \bar{A}\bar{B}\bar{D}(CE+1) + ABD(1+CE) + ABC\bar{D}E$$

$$f = \bar{A}BC\bar{D}E + \bar{A}\bar{B}\bar{D} + ABD + ABC\bar{D}E$$

$$f = BC\bar{D}E(A+\bar{A}) + ABD + \bar{A}\bar{B}\bar{D}$$

$$f = \underline{BC\bar{D}E + ABD + \bar{A}\bar{B}\bar{D}}$$

b) $f = \bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

Ans:

$$f = \bar{B}\bar{C}(1+\bar{A}) + ABD + \bar{A}C(1+\bar{D}) + A\bar{C}D$$

$$f = \bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D$$

$$f = \bar{B}\bar{C} + ABD + \bar{A}C + AB\bar{C}D + A\bar{B}\bar{C}D$$

$$f = \bar{B}\bar{C}(1+AD) + ABD(1+\bar{C}) + \bar{A}C$$

$$f = \underline{\bar{B}\bar{C} + ABD + \bar{A}C}$$

c) $f = \bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$

Ans:

$$f = \bar{A}\bar{C} + BC(1+\bar{A}) + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$f = \bar{A}\bar{C} + BC + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$f = \bar{A}\bar{C} + BC + A\bar{B} + ABC\bar{D} + A\bar{B}C\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$f = \bar{A}\bar{C} + BC(1+A\bar{D}) + A\bar{B}(1+C\bar{D}) + \bar{B}\bar{C}\bar{D}$$

$$f = \bar{A}\bar{C} + BC + A\bar{B} + \bar{B}\bar{C}\bar{D}$$

$$f = \bar{A}\bar{C} + BC + A\bar{B} + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$f = \bar{A}\bar{C}(1+\bar{B}\bar{D}) + BC + A\bar{B}(1+\bar{C}\bar{D})$$

$$f = \underline{\bar{A}\bar{C} + BC + A\bar{B}}$$

$$c.) f = \overline{x_1 x_3 x_4} + x_3 x_4 + \overline{x_1} \overline{x_2} x_4 + x_1 x_2 \overline{x_3} x_4 + x_2$$

ans:

$$f = x_2(1 + x_1 \overline{x_3} \overline{x_4}) + \overline{x_3} \overline{x_2} \overline{x_4} + x_3 x_4 + \overline{x_1} \overline{x_2} x_4$$

$$f = x_2(x_2 + \overline{x_1} \overline{x_2} x_4) + \overline{x_3} \overline{x_2} \overline{x_4} + x_3 x_4$$

$$f = x_2 + \overline{x_1} x_4 + \overline{x_1} \overline{x_3} \overline{x_4} + x_3 x_4$$

$$f = \overline{x_1}(x_4 + \overline{x_3} \overline{x_4}) + \overline{x_3} x_4 + x_2$$

$$f = \overline{x_1}(x_4 + \overline{x_3}) + x_3 x_4 + x_2$$

$$f = \overline{x_1} x_4 + \overline{x_1} \overline{x_3} + x_3 x_4 + x_2$$

$$f = [\overline{x_1} \overline{x_3} + x_3 x_4 + \overline{x_1} x_4] + x_2$$

$$f = \underline{\underline{\overline{x_1} \overline{x_3} + x_3 x_4 + x_2}}$$

3.) Which single gate can represent the functionality of given circuit?

Ans: By seeing the output of the 4 NAND gates before the final NAND gate we get

1.) $\overline{ABC}$

2.) $\overline{A\overline{B}C}$

3.) $\overline{A\overline{B}\overline{C}}$

4.) $\overline{A\overline{B}C}$

The final NAND gate performs a product of complements, which via DeMorgan's Law yields the sum-of-products:

$$Z = ABC + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}C$$

This equation is true whenever there is an odd number of 1's on the inputs.

$\therefore$ The gate this can be replaced by is 3-input XOR